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One-Class Classifier Performance: Comparing Majority versus Minority Class Training

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***Abstract*—Our study evaluates the impact of training *One-Class Classification* (OCC) algorithms on the majority class compared to training them on the minority class, using large and big datasets that are highly imbalanced. It is important to note that class availability can present a significant challenge in model training. In reality, there may be situations where one class is readily obtainable within a reasonable time frame while another class is not. Our task involves detecting instances of fraud in the Credit Fraud Detection Dataset and our Medicare dataset derived from Medicare Part D data and *List of Excluded Individuals and Entities* (LEIE) data. The Credit Card Fraud Detection Dataset has real-world transaction content as well as a significant class imbalance, making it suitable for use as a benchmark for credit card fraud detection. In addition, it is the only publicly available large data for credit card fraud analysis. Part D is big data, allowing researchers to analyze national trends and patterns in prescription drug usage and expenditures. The algorithms used in the study are One-Class *Gaussian Mixture Model* (GMM), *One-Class Adversarial Nets* (OCAN), and One-Class *Support Vector Machine* (SVM). Their performance is measured with the *Area Under the Precision-Recall Curve* (AUPRC) and *Area Under the Receiver Operating Characteristic Curve* (AUC). Our results indicate that OCC produces better results when models are trained on the majority class.**

***Index Terms*—one-class classification, credit card fraud, Medicare fraud, high class imbalance, big data**

1. Introduction

Binary classification is a well-established method in machine learning for addressing classification problems, where a model is trained on examples from two classes. However, factors such as class noise [1], class imbalance [2], and limited data [3] can considerably impact the performance of a binary classification model. To address these challenges and enhance the efficiency of binary classification models, researchers have suggested various strategies. It is vital for a binary classification model to exhibit high performance for both classes, not just the minority or positive class, which is often the focal point.

In certain cases, machine learning algorithms may struggle to identify patterns in data when an inadequate number of instances are present in one or both classes. Although it is crucial for practitioners to have a sufficient number of instances in the class of interest, this is not always achievable in practice. *One-Class Classification* (OCC), unlike binary classification, involves the training of examples from only one class. Two prevalent tasks in OCC are outlier detection [4] and novelty detection [5]. Although similar in concept, outlier detection and novelty detection have different objectives. Outlier detection identifies instances in a dataset that deviate significantly from the majority due to factors such as measurement errors, data entry errors, natural variability, and data diversity. Identifying and eliminating such outliers can improve a machine learning model's accuracy. Novelty detection, on the other hand, aims to recognize new, previously unseen instances that differ from the training data without the need for class labels. Novelty detection applications include intrusion detection, fraud detection, and surveillance.

In our investigation, we use two binary class datasets for fraud detection. The Credit Card Fraud Detection Dataset is an anonymized collection of financial transactions available on the Kaggle website [6]. Our Medicare Part D Dataset is a merger of Part D [7] data from the *Centers for Medicare and Medicaid Services* (CMS) website with the *List of Excluded Individuals and Entities* (LEIE) [8] data. The Credit Card Fraud Detection Dataset contains real-world transaction content, has a substantial class imbalance, and could potentially serve as a benchmark for credit card fraud detection. It is also the only large credit card fraud dataset available to the public. Part D is big data and contains information on prescription drug events for Medicare enrollees. The dataset provides insights into the types and quantities of medications prescribed, the prescribing healthcare providers, and the associated costs. This enables researchers to analyze trends and patterns in prescription drug usage and expenditures at the national level. The LEIE includes information on individuals and entities excluded from federal healthcare programs such as Medicare and Medicaid due to fraudulent or abusive activities.

In a dataset with binary class labels, class imbalance occurs when one class has significantly more instances than the other, resulting in a majority class (usually the negative class) and a minority class (typically the positive class). If the number of instances in the two classes varies greatly, as in the case of high class imbalance, it can influence the outcome of a machine learning experiment. High class imbalance can be defined as a minority-to-majority instance ratio between 1:100 and 1:10,000 [9].

It is important to underscore that class availability can present a significant challenge in model training. In reality, we may encounter situations where one class of data can be readily and efficiently collected within a reasonable time frame, while the procurement of another class might be faced with constraints. This discrepancy could be due to various

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factors such as data collection protocols, the nature of the event under investigation, and temporal variations. With regard to imbalanced big data, it would be beneficial if one could effectively train OCC algorithms on the minority class, as it is smaller and would require less computational resources. Our research evaluates the impact of training OCC algorithms on the majority class compared to training them on the minority class. To the best of our knowledge, we are the first to conduct such an investigation within the context of both the Credit Card Fraud Detection Dataset and a big data Medicare dataset. Our motivation stems from contrasting the rapid training of relatively fewer instances comprising the minority class against the time-consuming training of the numerically dominant majority class instances. The specific aim of this research contribution is to determine which training approach is more effective in identifying fraudulent instances in large data (credit card dataset) and big data (Medicare dataset). We use three one-class classifier learners for experimentation: One-Class *Support Vector Machine* (SVM) [10], One-Class *Gaussian Mixture Model* (GMM) [11], and *One-Class Adversarial Nets* (OCAN) [12]. The effectiveness of these classifiers is assessed using the *Area Under the Precision-Recall Curve* (AUPRC) [13] and *Area Under the Receiver Operating Characteristic Curve* (AUC) [14] metrics. For the class that is very easy to obtain data, we recommend having as many instances as possible based on the findings of this paper, assuming that these instances contribute to model performance.

The remainder of this paper is structured as follows: Section 2 reviews related work; Section 3 offers an overview of the Credit Card Fraud Detection Dataset and the Medicare Part D Dataset; Section 4 introduces the classifiers used; Section 5 details the training and testing methodology employed; Section 6 describes the metrics used to evaluate classification performance; Section 7 presents the results and discusses their implications; and Section 8 summarizes the primary contributions of this study and offers suggestions for future research.

2. Related Work

In this section, we examine OCC research that focuses on detecting fraudulent instances from the Credit Card Fraud Detection Dataset. There is no literature available on our Medicare Part D dataset with regard to OCC algorithms, due to the novelty of comparing OCC algorithms trained on the majority class with OCC algorithms trained on the minority class. For the related work that we discuss, it is worth noting that none have compared the impact of training learners on the majority class versus the minority class.

In their study, Li et al. [15] proposed a dynamic weighted entropy hybrid method to address class imbalance issues. They balanced the dataset using *Random Undersampling* (RUS) before training One-Class SVM and Isolation Forest. The evaluation was conducted using F1 score and AUPRC metrics. The authors trained anomaly detection models like Isolation Forest, One-Class SVM, and an autoencoder by incorporating a mix of minority and majority samples. The remaining samples formed an overlapping subset with a low imbalance ratio. This subset was employed alongside Random Forest, *Deep Neural Network*s (DNNs), and other nonlinear classifiers. Subsequently, Li et al. applied Dynamic Weighted Entropy, which strikes a balance between excluding minority class outliers and managing the imbalance ratio of the overlapping subset. Their study used the Credit Card Fraud Detection Dataset and private datasets. The authors tested a total of twelve models, six hybrid and six non-hybrid, with hybrid models demonstrating superior performance. The best results were obtained by a hybrid of an autoencoder and DNN, achieving F1 score and AUPRC values of 0.73 and 0.63, respectively. However, details about the train-test procedure for the Random Forest classifier were not provided. Additionally, the authors did not disclose the method they used for converting One-Class SVM predictive scores to probabilities. Obtaining AUPRC scores requires class probabilities, but One-Class SVM does not natively produce probability outputs.

For their research, Jeragh and AlSulaimi [16] explored using a hybrid model consisting of an autoencoder and OneClass SVM for OCC, comparing it to other models such as standalone autoencoder, standalone One-Class SVM, and an alternative hybrid model of an autoencoder and One-Class SVM. Evaluation metrics included Precision, Recall, F1 score, and *Geometric Mean* (G-mean). The distinction between the two hybrid models is that in the authors' proposed model, the One-Class SVM was trained on the mean squared error between input and output, while in the alternative approach, it was trained directly on the output. In the proposed model, the input was provided to the autoencoder, and then the input reconstruction error was computed and passed to the One-Class SVM. This model performed the best, with Precision, Recall, F1 score, and G-mean values of 0.938, 1, 0.9685, and 0.9998, respectively. We believe the paper's contribution could be better substantiated with the addition of a Related Work section.

Chandorkar [17] examined the performance of three anomaly detection methods, specifically Local Outlier Factor, Isolation Forest, and One-Class SVM, using metrics such as Precision, Recall, and F1 score. Only 10% of the Credit Card Fraud Detection dataset was utilized. Both Local Outlier Factor and Isolation Forest achieved Precision, Recall, and F1 scores of 1. However, Isolation Forest's Accuracy was 0.9974, while Local Outlier Factor's was 0.9966. It is worth noting that this paper lacks essential details and omits crucial information. For instance, the results for One-Class SVM are not provided. Additionally, it is important to recognize that Accuracy can be a misleading performance metric.

In a similar vein, Bodepudi [18] assessed the efficacy of the same classifiers (Local Outlier Factor, Isolation Forest, and One-Class SVM) as those employed by Chandorkar [17]. However, this study exclusively utilized the Accuracy metric for evaluation. Isolation Forest emerged as the leading classifier, with an Accuracy score of 0.9974. The sole reliance on the Accuracy metric presents a challenge, as it may

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obscure inadequate classification of the minority class in the presence of highly imbalanced data. Furthermore, the author's concise four-page paper lacks detailed information, such as the approach to data preprocessing.

Lastly, Ounacer et al. [19] conducted a comparative analysis of Isolation Forest's performance alongside two other anomaly detection classifiers, namely Local Outlier Factor and One-Class SVM. They also employed K-means clustering for anomaly detection. The chosen metric for evaluation was AUC. Ounacer et al. determined that the Isolation Forest model outperformed the others, achieving an AUC score of 0.9168. It is important to note that utilizing AUC as the sole metric in research can be deceptive [20] when classifying imbalanced data, a point elaborated upon in Section 6. Furthermore, the authors did not clarify whether all 30 features were used in the experiment, nor did they provide details regarding the method for converting One-Class SVM predictive scores into probabilities, which are necessary for obtaining AUC scores.

3. Datasets

This section is divided into two parts, one for each of the fraud datasets incorporated into our work. The Credit Card Fraud Detection Dataset is readily available online. Data for the Medicare Part D Dataset is also available online, but further processing was carried out to transform this Medicare data into our desired format.

A. Credit Card Fraud Detection Dataset

The Credit Card Fraud Detection Dataset was jointly published by Worldline and the *Université Libre de Bruxelles* (ULB). It contains transactions made by European credit cardholders and consists of 284,807 instances and 30 input features. Of these, 28 input features were transformed using *Principal Component Analysis* (PCA) [21]. Two features, "Time" and "Amount," were not transformed. "Time" denotes the duration in seconds between a transaction and the first transaction in the dataset, while "Amount" indicates the transaction value. Although "Amount" was normalized, "Time" was excluded from the analysis as it behaves like a unique identifier that could result in overfitting and impact the results.

The dataset has a binary label, where 1 represents a fraudulent transaction and 0 represents a non-fraudulent transaction. The dataset is highly imbalanced, with only 492 instances (0.172%) identified as fraudulent.

B. Medicare Part D Dataset

The Part D Dataset is based on physician claims for the Medicare Part D Prescription Drug Program. This data is available via download from the CMS. In the dataset, physicians are identified by their unique *National Provider Identifier* (NPI) and drugs are labeled with their brand and generic names. The dataset also includes information on average payments and charges, drug quantity prescribed, and medical specialty. The CMS has aggregated the data by NPI and drug name. Each row of the dataset includes an NPI, provider type, drug name, related information, and features such as gender. Each row represents a unique combination of NPI, provider type, and drug name for a physician. Aggregated records from less than 11 claims are not included in the dataset to protect the privacy of Medicare beneficiaries.

Our Medicare Part D dataset is sourced from prescription drug data spanning from 2013 to 2019. This dataset was annotated using LEIE data [8], which includes information about doctors who have committed fraudulent activities. The LEIE also offers additional information, such as the reason and date of exclusion. The establishment and oversight of the LEIE was carried out by the *Office of Inspector General* (OIG) [22], whose duty is to bar individuals and organizations from participating in healthcare programs funded by the federal government. However, it should be noted that only a portion of provider fraud in the U.S. is represented by NPI values within the LEIE.

The LEIE is joined to our processed Medicare Part D Dataset on NPI and year attributes, with physicians practicing within a year prior to their exclusion end year being labeled as fraudulent. The dataset has a binary label where a value of 1 indicates a physician who committed fraud and a value of 0 indicates a physician who has not committed fraud. Our final processed dataset consists of 5,344,106 instances and 82 features. This dataset is highly imbalanced, with only 3700 instances (0.0692%) identified as physicians who committed fraud.

4. Classifiers

In the experimentation phase, we used three machine learning algorithms, namely, One-Class GMM, One-Class SVM, and OCAN. All three of these are OCC learners that focus on training data associated with a single class while disregarding or rejecting data from other classes.

One-Class SVM is an algorithm that constructs a hypersphere in high-dimensional space. The aim is to encompass as many data points as possible from a single class. The center of the hypersphere is determined as the mean of the data points associated with the focused class, and the radius is set to enclose a specified percentage of data points. One-Class SVM is a popular choice due to its effectiveness in handling high-dimensional data and its ability to handle noisy data [23].

One-Class GMM is an algorithm designed to detect outliers that significantly deviate from a single class of data points. It is a generative model that represents the data distribution as a weighted sum of multiple Gaussian distributions. Each Gaussian component represents a cluster of similar data points. During training, the algorithm learns the parameters of the Gaussian distributions, such as mean and covariance, that best fit the focused class of data points. To classify a new data instance, One-Class GMM calculates the likelihood that the instance belongs to the focused class, which is determined by the probabilities of the instance being generated by each Gaussian component. If the likelihood falls below a certain threshold, the instance is considered an anomaly.

OCAN is an anomaly detection algorithm that utilizes adversarial training [24]. The algorithm aims to learn the

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underlying probability distribution of a single class of data points and differentiate between in-class and out-of-class samples. During training, OCAN simultaneously trains a generator network [25] to produce realistic in-class samples and a discriminator network to classify whether a given sample is in-class or not. The generator network generates in-class samples that are challenging for the discriminator network to distinguish from real in-class samples. The discriminator network tries to differentiate between real in-class samples and generated samples. Through this adversarial process, OCAN learns to identify the boundaries of the focused class and differentiate between in-class and out-of-class samples. To classify a new data instance, the algorithm computes the distance between the instance and the focused class in the feature space. If the distance falls below a certain threshold, the instance is considered in-class. Otherwise, it is considered an anomaly.

5. Training and Testing

For this study, the setup for the experiments involved utilizing a distributed computing platform comprised of nodes equipped with 16-core Intel Xeon CPUs, 256 GB RAM per CPU, and Nvidia V100 GPUs. Training and testing algorithms were implemented using the Python programming language. OCAN was used as a standalone library, while One-Class SVM and One-Class GMM were implemented from the Scikit-learn [26] library.

The data was divided into training and testing sets using an 80:20 ratio. When learners were trained on the majority class, instances from the minority class were excluded and vice-versa. The training phase employed the k-fold cross-validation method, with the model being trained on k-1 folds and tested on the remaining fold in each iteration to maximize data utilization. Stratification was applied to the cross-validation process to ensure proportional representation of each class across the folds. A value of k=5 was selected, with 4 folds allocated for training and 1 fold for testing.

6. Metrics

We define the simple performance metrics [27] shown in Table 1 as follows:

- *True Positive* (TP): the number of positive samples that are correctly identified as positive.
- *True Negative* (TN): the number of negative samples that are correctly identified as negative.
- *False Positive* (FP), also known as Type I error: the number of negative instances that are incorrectly identified as positive.
- *False Negative* (FN), also known as Type II error: the number of positive instances that are incorrectly identified as negative.

Other performance metrics can be derived from these fundamental metrics, as follows:

- *Recall*, also known as *True Positive Rate* (TPR) or sensitivity, can be calculated as TP/(TP + FN).

TABLE 1: Confusion Matrix

		Predicted Class	
		Positive	Negative
Actual Class	Positive	True Positive (TP)	False Negative (FN) (Type II error)
	Negative	False Positive (FP) (Type I error)	True Negative (TN)

- *Precision*, also known as positive predictive value, can be calculated as TP/(TP + FP).
- *False Alarm Rate*, also known as *False Positive Rate* (FPR), can be calculated as FP/(FP + TN).

To provide a better understanding of the challenges of evaluating machine learning models with highly imbalanced data, our research utilized two performance metrics, AUC and AUPRC.

The Area Under the Receiver Operating Characteristic *Receiver Operating Characteristic* (ROC) Curve, also known as AUC, is a measure used to evaluate a classifier's quality by summarizing the balance between TPR and FPR. AUC evaluates the performance of a classifier by considering all possible classification thresholds along the ROC curve. It provides a comprehensive assessment of the model's performance by consolidating it into a single value, making it easier to compare different classifiers. The AUC value ranges from 0 to 1, with a higher value indicating better classifier performance. A model that guesses randomly will have an AUC score of 0.5

AUPRC, which stands for Area Under the Precision-Recall Curve, measures the equilibrium between Precision and Recall by plotting these two metrics at different classification thresholds. The AUPRC score ranges from 0 to 1, with a higher value representing better classifier performance.

The definition of AUC suggests that a dataset with a high number of true negatives will result in a low number of false positives. However, the AUPRC definition does not consider the number of true negatives. Hence, for highly imbalanced datasets such as the Credit Card Fraud Detection Dataset, AUPRC provides a more accurate measure of false positives [28]. Consequently, in this study, AUPRC is considered more important than AUC.

Both AUC and AUPRC require class probabilities as inputs, which is not provided by the One-Class SVM by default, and hence calibration is necessary. For calibration, we used two different approaches, sigmoid calibration and isotonic regression. Sigmoid calibration adjusts the output predictive scores of the One-Class SVM algorithm using the Logistic Regression model, which allowed us to convert the output scores into class probabilities for the positive class. This method is based on Platt's work [29] and is more flexible than isotonic regression, as it can capture non-monotonic relationships between scores and probabilities. On the other hand, isotonic regression is a non-parametric method that fits a piecewise-constant function to the output scores of the classifier, which then maps the scores to probabilities in a monotonic manner. This method is computationally efficient

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and model-agnostic, which means it can be applied to any type of classifier. Zadrozny and Elkan [30] were the first to propose isotonic regression for classifier calibration.

In preliminary experiments, we found that the Scikit-learn implementation of One-Class GMM's predict_proba function for class membership probabilities was not suitable for calculating AUC or AUPRC. Therefore, we used the score_samples function of Scikit-learn's One-Class GMM. The score_samples function returns the log-likelihood of class membership for an instance. We then calibrated the output of score_samples with sigmoid calibration to compute class membership probabilities. Like One-Class GMM, the OCAN classifier natively produces output values that are probability estimates, and hence, calibration was not required.

7. Results and Discussion

Tables 2 and 3 present classification results for OCC algorithms trained on the majority class for the Credit Card Fraud Detection Dataset and the Medicare Part D Dataset, respectively. Each AUC or AUPRC score shown in Table 2 is the mean of 10 iterations of 5-fold cross-validation (50 performance scores). For Table 3, we note that results for One-Class SVM (Sigmoid) are not available due to long running times. In this table, we also note that scores for One-Class SVM (Isotonic) are the mean value of only 1 iteration of 5-fold cross-validation (5 performance scores) due to long running times. In this same table, scores for the One-Class GMM (Sigmoid) and OCAN models are the mean value of 10 iterations of 5-fold cross-validation (50 performance scores).

In the Credit Card Fraud Detection Dataset (Table 2), One-Class GMM (Sigmoid) demonstrated the best performance with the highest AUC and AUPRC scores of 0.9496 and 0.4971, respectively. On the other hand, One-Class SVM (Sigmoid) had the lowest AUC score of 0.9084, and OCAN had the lowest AUPRC score of 0.3471.

For the Medicare Part D Dataset (Table 3), One-Class GMM (Sigmoid) once again achieved the highest AUC and AUPRC scores of 0.7289 and 0.1481, respectively. In contrast, One-Class SVM (Isotonic) had the lowest AUC score of 0.5127, and OCAN showed the lowest AUPRC score of 0.0015.

TABLE 2: Credit Card Fraud Detection Dataset - Scores based on Majority Class

Classifier	AUC	AUPRC
One-Class GMM (Sigmoid)	0.9496	0.4971
OCAN	0.9409	0.3471
One-Class SVM (Isotonic)	0.9091	0.4165
One-Class SVM (Sigmoid)	0.9084	0.3775

TABLE 3: Medicare Part D Dataset - Scores based on Majority Class

Classifier	AUC	AUPRC
One-Class GMM (Sigmoid)	0.7289	0.1481
OCAN	0.5533	0.0015
One-Class SVM (Isotonic)	0.5127	0.0031

Tables 4 and 5 present classification results for OCC algorithms trained on the minority class for the Credit Card Fraud Detection Dataset and the Medicare Part D Dataset, respectively. Each AUC or AUPRC score shown in both tables is the mean of 50 performance scores.

In the Credit Card Fraud Detection Dataset (Table 4), One-Class GMM (Sigmoid) outperformed other models, achieving the highest AUC and AUPRC scores of 0.7382 and 0.2205, respectively. On the other hand, OCAN exhibited the lowest performance, with the lowest AUC and AUPRC scores of 0.4332 and 0.0172, respectively.

For the Medicare Part D Dataset (Table 5), One-Class SVM (Isotonic) achieved the highest AUC and AUPRC scores of 0.6082 and 0.0042, respectively. In contrast, One-Class GMM (Sigmoid) had the lowest AUC and AUPRC scores of 0.4739 and 0.0007, respectively.

TABLE 4: Credit Card Fraud Detection Dataset - Scores based on Minority Class

Classifier	AUC	AUPRC
One-Class GMM (Sigmoid)	0.7382	0.2205
OCAN	0.4332	0.0172
One-Class SVM (Isotonic)	0.5603	0.0334
One-Class SVM (Sigmoid)	0.5718	0.0184

TABLE 5: Medicare Part D Dataset - Scores based on Minority Class

Classifier	AUC	AUPRC
One-Class GMM (Sigmoid)	0.4739	0.0007
OCAN	0.4819	0.0009
One-Class SVM (Isotonic)	0.6082	0.0042
One-Class SVM Sigmoid)	0.6072	0.0017

As previously stated, this study prioritizes AUPRC scores as the primary performance metric in evaluating the OCC models. For the Credit Card Fraud Detection Dataset, all the learners trained on the majority class (Table 2) have AUPRC scores greater than the learners trained on the minority class (Table 4). One-Class GMM (Sigmoid) is the top performer with regard to learners trained on either class. For the Medicare Part D Dataset, One-Class GMM (Sigmoid) has a considerably higher score, when trained on the majority class, than any of the other learners trained on the majority class (Table 3) or minority class (Table 5). The poor performance of the other models trained on either the majority or minority class leads to the conclusion that One-Class GMM (Sigmoid), when trained on the majority class, is the only suitable model in this study for classifying the Medicare Part D Dataset.

The results of our research suggest that OCC tends to yield better results when models are trained on the majority class. However, we caution that the results are limited to the two datasets analyzed, and further research is needed to determine if these observations generalize to other datasets.

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8. Conclusion

There can be circumstances where data from one class is easily accessible and can be gathered in a practical time frame, while data from another class may not be as straightforward to acquire. Within the context of imbalanced big data, it would be advantageous if one could effectively train OCC algorithms on the minority class, as it is smaller and would require less computational resources. Our novel investigation compares the efficiency of rapid, smaller scale training against time-intensive, larger scale training using the Credit Card Fraud Detection and Medicare datasets. The objective is to identify which training method best detects fraud in these large and big data contexts.

We employed three OCC algorithms and evaluated their performance. The findings suggest that OCC models perform better when they are trained using data from the majority class. Additionally, the results indicate that One-Class GMM (Sigmoid) is the top performer overall when trained on the majority class for both datasets under consideration. This information could be valuable for researchers and practitioners working with large and big highly imbalanced datasets, as it provides insight into the potential benefits of training one-class classification models using majority class data. It also highlights the effectiveness of the One-Class GMM (Sigmoid) algorithm in these settings. According to the findings of this study, for a class where data instances can be easily collected, it may be beneficial to gather as many instances as feasible.

Our research lays the groundwork for further exploration in the domain of credit card fraud detection and Medicare fraud detection. The approach we have taken can be applied to assess OCC models with big data from other application domains.

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